

Shafagh Rastegari

Bologna, Italy | shafaghrastegari@gmail.com | (+39)351 734 4192 | [linkedin.com/in/shafagh-rastegari-411767223](https://www.linkedin.com/in/shafagh-rastegari-411767223)
github.com/ShafaghRastegari

About Me

Second-year AI Master student with a Computer Engineering background, specializing in Deep Learning, Machine Learning, NLP, and Large Language Models (LLM). Project experience includes fine-tuning a model to achieve a top-5 team ranking in an NLP challenge, developing an end-to-end LLM pipeline to remove biased reasoning, and engineering a Dockerized MIP optimization solution using PuLP. Eager to apply AI theory and advanced modeling skills to real-world projects in an AI/ML role.

Education

University of Bologna	Bologna, Italy
Master of Science in Computer Science, Artificial Intelligence	Sep 2023 – Current
Iran University of Science and Technology	Tehran, Iran
Bachelor of Science in Computer Engineering	Sep 2017 – Oct 2022

Experience

Front-end Developer	June 2021 – March 2022
Targoman Intelligent Processing Company	Tehran, Iran
• Developed an admin and user panel websites using Vue.js and Typescript and connect it to the backend.	

Projects

Cifar10 Image Separation <i>Deep Learning</i>	Repository Link
• A Deep Learning model that takes as input an image created by averaging two random samples from CIFAR-10 dataset and is tasked with predicting the categories of the two components using Convolutional Neural Network.	
• Tools Used: Python, Tensorflow, Keras, Numpy, Matplotlib	
Bias Mitigation in Reasoning LLMs with Multi-Judge Pipeline <i>LLM, Reasonable AI</i>	Repository Link
• A pipeline to detect and remove biased reasoning steps of Chain-of-Thought in reasoning LLMs to reducing stereotype propagation without compromising accuracy, benchmarked model performance across bias score and accuracy on BBQ (Bias Benchmark for QA) and MBBQ (Multilingual BBQ) datasets.	
• Tools Used: Python, LangChain	
Multiple Couriers Planning Problem <i>Optimization</i>	Repository Link
• Implemented a Mixed-Integer Programming (MIP) using PuLP to efficiently solve courier planning problem, and Dockerize it as combinatorial decision making and optimization approach.	
• Containerized the solution via Docker, guaranteeing the execution across different systems.	
• Tools Used: Python, MIP, PuLP, Docker	
Subjectivity in News Articles <i>NLP, LLM</i>	Repository Link
• Fine-tune mDeBERTaV3-base model for 5 different languages in 3 different settings (Multilingual, Monolingual, and zero-shot) for a binary classification task in order to detect the Subjectivity or Objectivity of the news sentences a part of this challenge. Our results placed us among the top 5 teams in almost all languages.	

Technical Skills

Languages: Python, C++, Prolog, Vue.js, React, JavaScript, TypeScript

Libraries: LangChain, TensorFlow, Keras, scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, PuLP

Technologies: Git, Docker, Latex

Languages

Persian: native, **English:** proficient, **Italian:** beginner